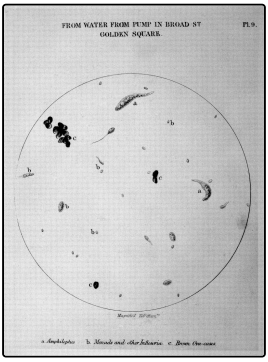
Water & Sanitation Through Time

KNOWLEDGE RETRIEVAL & HOMEWORK COMPANION • 30 HIGH-YIELD QUESTIONS

Pupil Name: _____ **Class:** _____ **Teacher:** _____



Microscopic examination of contaminated water from the Broad Street pump during the 1854 Soho cholera epidemic • Dr John Snow proved waterborne transmission, revolutionising epidemiology.

LESSON ENQUIRY FOCUS	HW DUE DATE	1. SOLO (WRITTEN)	2. PARTNER (VERBAL)	PARENT / TEACHER SIGN
Enquiry 1: Roman Public Health & Conduits Stone aqueducts, Housesteads fort communal latrines, public baths (hypocaust, strigil), Cloaca Maxima & post-Roman collapse		/ 6	/ 6	
Enquiry 2: Medieval Sanitation & Black Death Town cesspits, gongfermors, monastic sanitation (Canterbury waterworks plan), 1348 Black Death arrival & social upheaval		/ 6	/ 6	
Enquiry 3: Early Modern Filth & 1665 Plague Urban crowding, privy pits, Sir John Harington's flush toilet (1596), 1665 Great Plague, Plague Orders & 1666 Great Fire		/ 6	/ 6	
Enquiry 4: Industrial Slums & Cholera Urban population explosion, back-to-back housing, 1831 cholera arrival, Edwin Chadwick's 1842 Report & Dr John Snow's Broad Street map		/ 6	/ 6	
Enquiry 5: The Great Stink & Germ Theory Summer 1858 Great Stink, Joseph Bazalgette's 82-mile sewer network, Thames Embankment, Louis Pasteur's 1861 Germ Theory & 1875 Act		/ 6	/ 6	

HOW TO COMPLETE YOUR RETRIEVAL CYCLES:

1. SOLO (WRITTEN): Complete your 6 write-in lines from memory without looking. Check your answers in green pen using the Department Marking Bank on Pages 7–8.

2. PARTNER (VERBAL): Hand your booklet to a **PARENT, CARER, FAMILY MEMBER, OR STUDY BUDDY** to test you out loud from the Marking Bank until you hit 6/6!

Scan for Interactive Digital Quizzes & Audio Lessons:
Access online self-marking retrieval flashcards, diagnostic feedback, and audio narration at the-history-revision-hub.netlify.app.

Enquiry 1: Roman Public Health & Conduits

How much progress did the Romans make in public health and municipal infrastructure?

QUESTIONS 1–6

100 BC Roman Aqueduct Building → 122 AD Hadrian's Wall Latrines → 200 AD Roman Bathhouses in Bri… → 300 AD Cloaca Maxima Flushes … → 410 AD Legions Leave; Systems …

1. ► **What technology did Roman engineers construct to transport millions of gallons of fresh water into towns?**

Fact: _____

Explanation: _____

2. ► **Which Roman fort on Hadrian's Wall contains Britain's best-preserved communal latrines with flowing sewers?**

Fact: _____

Explanation: _____

3. ►► **What curved bronze tool did bathers use with olive oil to scrape dirt and sweat from their bodies in Roman baths?**

Fact: _____

Explanation: _____

4. ►► **Did Roman emperors build public health systems because they understood bacterial germs?**

Fact: _____

Explanation: _____

5. ►►► **How did Roman emperors and military governors use clean water and bathhouses as a political weapon?**

Fact: _____

Explanation: _____

6. ►►► **Why did Roman public health infrastructure collapse after the legions left Britain in 410 AD?**

Fact: _____

Explanation: _____

KEY TERM VAULT: | Aqueduct | Cloaca Maxima | Housesteads | Hypocaust | Strigil | Miasma

Enquiry 2: Medieval Sanitation & Black Death

Why did public health decline during the Middle Ages, and how did society respond to plague?

QUESTIONS 7–12

1165 Canterbury Waterworks ... → **1237** The Great Conduit Built → **1345** London Dung Dumping B... → **1348** Black Death Strikes Britain → **1349** King Orders Streets Clea...

7. ► **What profession was paid to empty human waste from town cesspits in medieval England?**

Fact: _____

Explanation: _____

8. ► **Why were medieval monasteries like Canterbury Cathedral much cleaner and healthier than ordinary towns?**

Fact: _____

Explanation: _____

9. ►► **When did the Black Death arrive in England, and what percentage of the population did it kill?**

Fact: _____

Explanation: _____

10. ►► **How did medieval town councils attempt to prevent the spread of the Black Death?**

Fact: _____

Explanation: _____

11. ►►► **Why did medieval towns like London build "The Great Conduit" in 1237?**

Fact: _____

Explanation: _____

12. ►►► **How did the Black Death fundamentally alter the socio-economic balance of power in medieval England?**

Fact: _____

Explanation: _____

KEY TERM VAULT: | Cesspit | Gongfermor | Monastic Waterworks | Great Conduit | Black Death 1348 | Bubonic Plague

Enquiry 3: Early Modern Filth & 1665 Plague

To what extent did towns become filthier, and how did London react to the 1665 Great Plague?

QUESTIONS 13–18

1596 Harington Invents Water … → **1609** New River Brings Spring … → **1665** Great Plague Kills 100,000 → **June 1665** Lord Mayor's Plague Ord… → **Sept 1666** Great Fire Destroys Slums

13. ► **How many people died in London during the Great Plague of 1665?**

Fact: _____

Explanation: _____

14. ► **What marking was painted on the doors of infected houses during the 1665 Great Plague?**

Fact: _____

Explanation: _____

15. ►► **Who were the "Searchers of the Dead" appointed by parish officials during the Great Plague?**

Fact: _____

Explanation: _____

16. ►► **Why did wealthy Londoners, including King Charles II and physicians, survive the plague at higher rates?**

Fact: _____

Explanation: _____

17. ►►► **Why was Sir John Harington's 1596 invention of the flush water closet ignored for nearly two centuries?**

Fact: _____

Explanation: _____

18. ►►► **How did the Great Fire of London in September 1666 unexpectedly improve public health?**

Fact: _____

Explanation: _____

KEY TERM VAULT: | Great Plague 1665 | Plague Orders | Searchers of Dead | Red Cross | Pest House | Great Fire 1666

Enquiry 4: Industrial Slums & Cholera

How did industrialisation cause a public health crisis, and how was cholera tracked?

QUESTIONS 19–24

1831 First Cholera Outbreak → **1842** Chadwick's Sanitary Rep… → **1848** First Public Health Act → **1854** Broad Street Pump Outbr… → **Sept 1854** Pump Handle Removed

19. What terrifying new waterborne disease struck Britain for the first time in 1831?

Fact: _____

Explanation: _____

20. What housing design became common in industrial boom towns like Leeds and Manchester?

Fact: _____

Explanation: _____

21. What did Edwin Chadwick's landmark 1842 Sanitary Report argue was the true cause of working-class poverty?

Fact: _____

Explanation: _____

22. Why was the 1848 Public Health Act largely ineffective across most British towns?

Fact: _____

Explanation: _____

23. How did Dr John Snow prove that cholera was transmitted by contaminated water in Soho in 1854?

Fact: _____

Explanation: _____

24. Why did the scientific establishment reject Dr John Snow's waterborne cholera discovery in 1854?

Fact: _____

Explanation: _____

KEY TERM VAULT: Cholera | Back-to-Backs | Edwin Chadwick | 1848 Public Health Act | Dr John Snow | Broad Street Pump

Enquiry 5: The Great Stink & Germ Theory

Why did the Great Stink clean up streets, and how did Germ Theory conquer waterborne disease?

QUESTIONS 25–30

1858 The Great Stink Paralyze... → 1859–65 Bazalgette Builds Sewers → 1861 Pasteur Publishes Germ ... → 1870 Victoria Embankment Op... → 1875 Public Health Act (Comp...

25. ► **What environmental crisis in the summer of 1858 finally forced Parliament to fund London's sewers?**

Fact: _____

Explanation: _____

26. ► **Which master civil engineer was appointed to design and build London's revolutionary sewer network?**

Fact: _____

Explanation: _____

27. ►► **How did Joseph Bazalgette's sewer design transform London's riverfront geography?**

Fact: _____

Explanation: _____

28. ►► **What revolutionary scientific breakthrough did Louis Pasteur publish in 1861?**

Fact: _____

Explanation: _____

29. ►►► **How did Robert Koch's laboratory research in the 1870s and 1880s provide definitive proof for Snow's findings?**

Fact: _____

Explanation: _____

30. ►►► **Why is the Public Health Act 1875 considered the decisive turning point in British public health history?**

Fact: _____

Explanation: _____

KEY TERM VAULT: | The Great Stink | Joseph Bazalgette | Intercepting Sewers | Thames Embankment | Germ Theory (Pasteur) | 1875 Public Health Act

Department Marking Bank • Part 1

Green-Pen Verification Key • Enquiries 1 to 3 (Questions 1 to 18)

ENQUIRIES 1–3

ENQUIRY 1: ROMAN PUBLIC HEALTH & CONDUITS

1. ► **Stone aqueducts and conduits using gravity** — *Romans engineered gently sloping stone channels that carried clean river or spring water across valleys directly into public fountains.* [✓] [✗]

2. ► **Housesteads Fort on Hadrian's Wall** — *Soldiers sat side-by-side on stone benches while rainwater and runoff flowed underneath to flush waste continuously into drainage ditches.* [✓] [✗]

3. ►► **A strigil** — *Romans did not use soap; instead, they rubbed olive oil into their skin and scraped off dirt before moving between heated bath chambers.* [✓] [✗]

4. ►► **No, they believed in balancing the Four Humours and avoiding miasma (bad air)** — *Romans associated foul smells and stagnant water with disease, but had zero knowledge of microscopic pathogens.* [✓] [✗]

5. ►►► **By displaying imperial wealth and keeping colonial subjects content and civilised** — *Free or cheap baths demonstrated Roman supremacy and cultural luxury, winning the loyalty of conquered provincial elites.* [✓] [✗]

6. ►►► **Central imperial taxation, engineering expertise, and municipal maintenance collapsed** — *Without central administration, aqueducts fell into ruin and Anglo-Saxon settlers preferred rural wooden homesteads.* [✓] [✗]

ENQUIRY 2: MEDIEVAL SANITATION & BLACK DEATH

7. ► **Gongfermors (or gong scourers)** — *Working only at night, gongfermors shoveled sewage from underground pits and carted it beyond town walls to sell as farm fertilizer.* [✓] [✗]

8. ► **Piped spring water, settling tanks, flushing latrines, and isolated infirmaries** — *Monks had the wealth and literacy to study Roman engineering, valuing bodily cleanliness as a religious duty to God.* [✓] [✗]

9. ►► **1348, killing an estimated 30% to 50% of the entire British population** — *Carried by fleas on black rats aboard trading ships, the plague decimated villages and towns within eighteen months.* [✓] [✗]

10. ►► **By lighting bonfires to clear bad air, burying dead in deep pits, and isolating houses** — *Believing in miasma and divine punishment, officials banned public gatherings and ordered butchers to dispose of animal offal.* [✓] [✗]

11. ►►► **To pipe clean spring water from Tyburn into public lead cisterns at Cheapside** — *As London's population grew, local wells and the River Thames became polluted by dye houses, tanneries, and cesspits.* [✓] [✗]

12. ►►► **By causing labour shortages, enabling peasants to demand higher wages and freedom from serfdom** — *Landlords competed desperately for scarce agricultural workers, fatally weakening feudalism and paving the way for the 1381 Peasants' Revolt.* [✓] [✗]

ENQUIRY 3: EARLY MODERN FILTH & 1665 PLAGUE

13. ► **Approximately 100,000 people, roughly one-quarter of London's population** — *The outbreak was the worst epidemic in England since the Black Death, striking poorest crowded parish slums like St Giles hardest.* [✓] [✗]

14. ► **A red cross, with the words "Lord have mercy upon us"** — *Under the Lord Mayor's Orders, watchmen were stationed outside day and night to lock families inside for 40 days of quarantine.* [✓] [✗]

15. ►► **Elderly parish women hired to inspect corpses and certify causes of death** — *Searchers reported causes of death to the parish clerk to compile the weekly "Bills of Mortality", tracking the epidemic's spread.* [✓] [✗]

16. ►► **Because they fled London to country estates, leaving poor residents trapped in slums** — *Those who could afford carriages evacuated the capital, leaving working-class citizens to bear the brunt of infection and quarantine.* [✓] [✗]

17. ►►► **Houses lacked running water pressure, and towns had no sewer pipes to carry waste away** — *Without municipal plumbing, a water closet simply flushed waste into an indoor cesspool beneath the kitchen floor.* [✓] [✗]

18. ►►► **By destroying overcrowded, rat-infested timber slums and replacing them with wider brick streets** — *Rebuilding regulations mandated wider thoroughfares and brick construction, significantly reducing black rat nesting sites in the city center.* [✓] [✗]

Department Marking Bank • Part 2

Green-Pen Verification Key • Enquiries 4 to 5 (Questions 19 to 30)

ENQUIRIES 4–5

ENQUIRY 4: INDUSTRIAL SLUMS & CHOLERA

19. ► **Asiatic Cholera, killing victims within hours through catastrophic dehydration** — *Originating in India, cholera spread along trade routes; victims suffered violent diarrhea, their skin turning blue from blood thickening.* [✓] [✗]

20. ► **Back-to-back terrace houses with no gardens, through-ventilation, or private toilets** — *Landlords built rows touching at the back and sides to maximize rental yield, forcing hundreds of tenants to share outdoor privies.* [✓] [✗]

21. ►► **Filth, polluted water, and disease caused poverty, not moral weakness or laziness** — *Chadwick demonstrated that poor sanitation killed workers young, leaving families on poor relief, costing taxpayers far more than sewers.* [✓] [✗]

22. ►► **It was permissive, allowing local town councils to refuse to spend money on clean water** — *Dominated by wealthy ratepayers adhering to laissez-faire, councils resented state interference and refused to levy taxes to fund drainage.* [✓] [✗]

23. ►►► **By mapping cholera deaths and proving victims drank water from the Broad Street pump** — *Snow showed that 500 deaths clustered around one pump, while nearby brewery workers who drank only beer were completely unaffected.* [✓] [✗]

24. ►►► **Doctors and the Board of Health clung firmly to traditional Miasma Theory (bad air)** — *Prominent figures like William Farr argued disease was inhaled from foul smells; Snow's theory was only accepted decades later after Germ Theory.* [✓] [✗]

ENQUIRY 5: THE GREAT STINK & GERM THEORY

25. ► **The Great Stink of the River Thames** — *A heatwave baked millions of tons of raw human sewage in the Thames, creating an unbearable stench that disrupted parliamentary debates.* [✓] [✗]

26. ► **Sir Joseph Bazalgette** — *Bazalgette engineered 82 miles of massive underground brick intercepting sewers, transporting waste eastward beyond the city to tidal outfalls.* [✓] [✗]

27. ►► **By constructing the Thames Embankment (Victoria, Albert, Chelsea) over the sewer lines** — *Bazalgette reclaimed 52 acres of riverside mudflats, enclosing low-level sewers, underground rail lines, and gas mains beneath wide roads.* [✓] [✗]

28. ►► **The Germ Theory of disease, proving microscopic pathogens cause decay and illness** — *Pasteur proved that microorganisms in the air caused fermentation, disproving spontaneous generation and providing the scientific basis for health.* [✓] [✗]

29. ►►► **By isolating and photographing the specific comma-shaped bacterium (Vibrio cholerae)** — *Koch used agar plates and industrial dyes to identify the exact microbial pathogen contaminating drinking water, finally ending Miasma Theory.* [✓] [✗]

30. ►►► **It made clean water, sewer connections, and rubbish disposal compulsory for all councils** — *The Act dismantled Victorian laissez-faire ideology, legally obliging every municipal authority to appoint medical officers and inspect housing.* [✓] [✗]

UNIT CONCEPTUAL FOUNDATIONS • 4 CORE ANCHORS

MIASMA THEORY: The ancient belief that diseases like the plague and cholera were caused by inhaling foul-smelling air from rotting organic filth.

LAISSEZ-FAIRE: The political philosophy that governments should not interfere in private enterprise or tax ratepayers for public health infrastructure.

EPIDEMIOLOGY (DR SNOW): The scientific study of disease patterns, pioneered in 1854 by Dr John Snow mapping cholera deaths to the Broad Street pump.

GERM THEORY & 1875 ACT: Louis Pasteur's discovery of microscopic pathogens, which compelled parliament to pass the mandatory 1875 Public Health Act.